

DRIVE-C: A Controlled Corruption Dataset for Autonomous Driving

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Abstract

DRIVE-C is a controlled corruption dataset designed to evaluate visual perception robustness in autonomous driving systems. It is built from real-world forward-facing driving videos collected across daytime, nighttime, urban, rural, freeway, and parking environments. Clean clips are anonymized via localized face and license plate blurring, then transformed with physics-inspired synthetic degradations.

The dataset contains 10 clean clips and 600 corrupted clips spanning 12 camera degradation types across five severity levels, with per-clip metadata and Global Sensor Health Index (GSHI) annotations. DRIVE-C supports robustness benchmarking, degradation-aware modeling, uncertainty estimation, out-of-distribution (OOD) detection, and sensor health monitoring for Advanced Driver Assistance Systems (ADAS). By providing pixel-aligned clean and degraded video clips with fully reproducible corruption parameters, DRIVE-C offers a structured testbed for studying perception reliability under controlled camera degradation.

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Data type/location: Video dataset with clean and synthetically corrupted clips, associated metadata, and GSHI annotations; collected in Michigan, United States; hosted on Zenodo.

Keywords: autonomous driving, camera degradation, corruption dataset, driving perception, robustness, sensor health, uncertainty estimation.

BACKGROUND

Camera-based perception is a core component of Advanced Driver Assistance Systems (ADAS) and autonomous driving, yet its performance is highly sensitive to image degradations encountered in real-world operation. These include low-light conditions, glare, motion blur, sensor noise, adverse weather, compression artifacts, and partial occlusions. While widely used driving datasets such as KITTI [1], nuScenes [2], and Waymo [3] support perception tasks under natural conditions, they do not provide systematic control over degradation types and severity. Conversely, corruption benchmarks such as

ImageNet-C [4] are typically derived from web imagery and do not preserve the viewpoint, geometry, and temporal characteristics of onboard automotive cameras.

DRIVE-C is designed to bridge this gap by combining real-world driving footage with controlled, reproducible degradations. Clean video clips are drawn from forward-facing vehicle-mounted camera recordings across diverse environments, including urban, suburban, freeway, parking, daytime, and nighttime scenarios. A structured set of degradations is then generated using physics-inspired models at multiple severity levels, enabling controlled robustness evaluation while retaining realistic scene characteristics.

The dataset is intended to support a broad range of data-centric research tasks. These include robustness benchmarking of perception models under domain shift, degradation-aware training and fine-tuning, out-of-distribution (OOD) detection, uncertainty calibration, failure mode analysis, and sensor health monitoring for safety-critical ADAS applications. The availability of aligned clean and degraded clip pairs with structured metadata enables reproducible evaluation and direct comparison across degradation conditions and model architectures. Researchers developing or evaluating object detectors, lane estimators, depth predictors, or semantic segmentation models under realistic sensor degradation scenarios will find DRIVE-C directly applicable.

Recent dataset descriptor publications further emphasize the importance of structured, well-documented datasets for reproducible research [5]. Comparable corruption datasets in adjacent domains, such as ImageNet-C, have become standard benchmarks within their communities; DRIVE-C aims to serve a similar role for automotive camera perception.

This dataset has not previously been used in any publication. The current release covers 10 capture scenarios and 12 corruption types, each applied at 5 severity levels. Detailed dataset statistics are provided in the Records and Storage section. The DRIVE-C dataset is publicly available on Zenodo [6].



Fig. 1: Teaser of DRIVE-C. A clean forward-facing driving clip is paired with controlled corrupted variants across multiple camera degradation types. DRIVE-C contains 10 scenarios, 12 corruption types, 5 severity levels, 610 clips, 78,080 frames, and metadata including severity and Global Sensor Health Index (GSHI) labels.

COLLECTION METHODS AND DESIGN

Data Acquisition Hardware and Camera Configuration

DRIVE-C is collected using a forward-facing action camera mounted inside a passenger vehicle to approximate the viewpoint of a windshield-mounted automotive perception sensor. The capture platform is a DJI Osmo Action 4 [7], installed in a fixed forward-facing configuration to ensure viewpoint consistency across recording sessions. The camera records natively at 3840×2160 (4K UHD) and 30 fps; all clips are downsampled to 1280×720 (HD) for release to reduce storage overhead while retaining sufficient spatial resolution for perception tasks.

Camera parameters are manually controlled where possible to reduce inter-session variability. For nighttime driving, representative settings include a shutter speed of $1/40$ s, ISO 1600, and white balance of 4000 K. For daytime driving, lower ISO values and faster shutter speeds are used, with fixed white balance to avoid frame-to-frame color variation. Cloudy and transitional lighting conditions are adjusted to reduce highlight clipping and washed-out sky regions. In-camera low-light enhancement is disabled to preserve native sensor characteristics.

Capture Scenarios

Data are collected from real-world driving sessions across diverse environments in Michigan, United States, on April 1 and April 5, 2026. Environments include urban streets, suburban roads, rural roads, multi-lane freeways, and a parking lot. Many recordings contain continuous transitions between scene types. As a result, DRIVE-C

Table 1: Per-scenario metadata for the 10 DRIVE-C capture scenarios.

ID	Scene Type	Weather	Time	Traffic	Frames
S01	Urban	Cloudy	Day	Moderate	128
S02	Freeway	Cloudy	Day	Moderate	128
S03	Urban	Sunny	Day	Moderate	128
S04	Rural	Cloudy	Day	Moderate	128
S05	Parking lot	Sunny	Day	Moderate	128
S06	Urban	Cloudy	Night	Moderate	128
S07	Urban	Cloudy	Night	Moderate	128
S08	Urban	Cloudy	Night	None	128
S09	Urban	Sunny	Day	Low	128
S10	Rural	Cloudy	Day	Moderate	128

does not rely on rigid per-frame scene labels as primary metadata fields; instead, each clip is associated with a high-level scene type and capture condition metadata, enabling flexible downstream labeling.

The dataset comprises 10 scenarios (S01–S10), each a fixed 128-frame clip (approximately 4.3 s at 30 fps) drawn from a distinct source video. The scenarios span three lighting and weather regimes:

- **Daytime, clear/sunny** — scenarios S03, S05, S09 (scene types: urban, parking lot, urban),
- **Daytime, cloudy** — scenarios S01, S02, S04, S10 (scene types: urban, freeway, rural, rural),
- **Nighttime, cloudy** — scenarios S06, S07, S08 (scene types: urban, urban, urban).

Table 1 summarizes the per-scenario metadata.

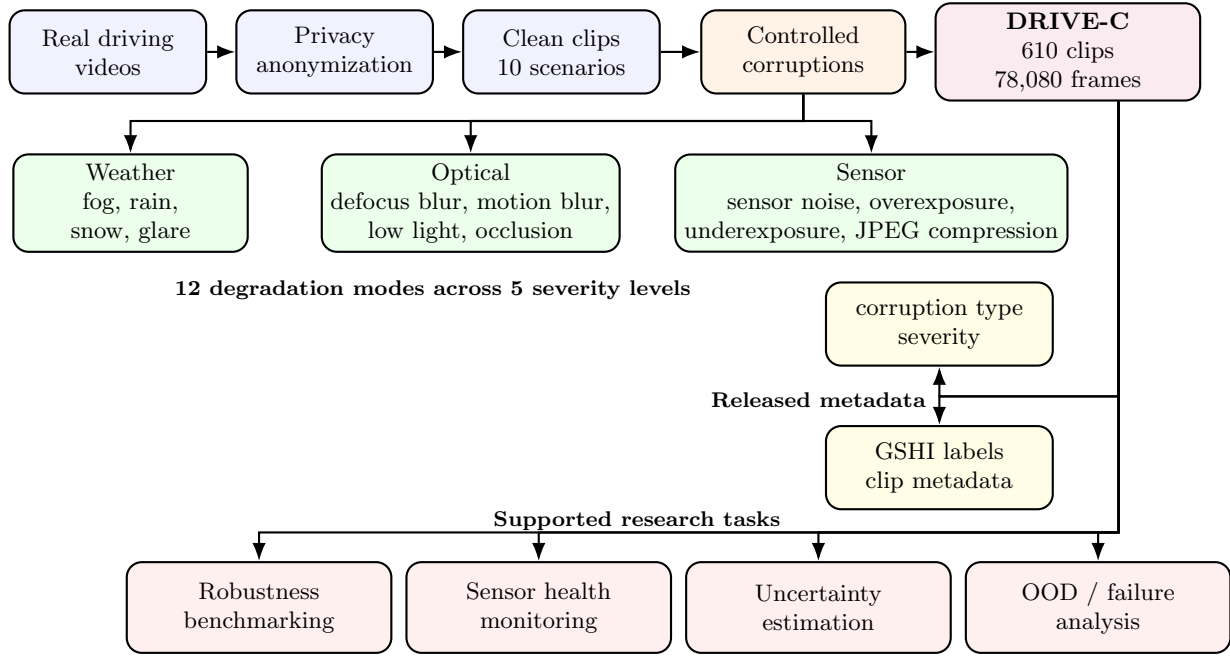


Fig. 2: Overview of DRIVE-C. Real forward-facing driving videos are anonymized and converted into clean clips, then transformed using controlled camera corruptions across weather, optical, and sensor/pipeline degradation families. Each corrupted clip includes severity metadata and GSHI labels, enabling robustness benchmarking, uncertainty estimation, and sensor health monitoring for autonomous driving perception.

Data Processing Pipeline

The dataset is constructed using a reproducible multi-stage pipeline:

1. Record forward-facing driving video at 4K/30 fps using a fixed vehicle-mounted camera.
2. Select a 128-frame clip from each source video, down-sampled to 1280×720 .
3. Organize clips by scenario identifier in a structured directory hierarchy.
4. Apply automated anonymization to faces and license plates.
5. Record processing provenance in per-frame and per-clip metadata files.
6. Perform quality review to confirm scene coverage and lighting representativeness.
7. Apply 12 controlled corruption types at 5 severity levels using a reproducible degradation pipeline to produce the final corrupted clip set.

This design separates real-world data acquisition from synthetic corruption generation and preserves full traceability from each corrupted clip back to its clean source clip and originating source video.

Anonymization Strategy

To protect privacy in real-world driving imagery, DRIVE-C employs an automated anonymization pipeline based on pretrained YOLO-family detectors [8] executed with GPU inference. The anonymization process includes:

- localized elliptical blurring for detected faces,
- localized rectangular blurring for detected license plates,
- optional fallback anonymization based on broader person or vehicle detections when primary detectors yield low confidence.

Face blurring is applied using elliptical masks, and license plate blurring is tightly constrained to detected bounding regions rather than applying coarse masking, preserving surrounding scene context for downstream perception tasks.

Controlled Corruption Design

DRIVE-C follows a two-stage design: a clean anonymized base clip set and a derived corruption clip set. Corruptions are applied exclusively to clean clips to ensure that privacy processing does not interfere with degradation severity or introduce confounding artifacts.

The dataset includes 12 corruption types spanning three degradation families, as listed in Table 2. The corruption models are adapted from the benchmark framework of Hendrycks and Dietterich [4] and implemented using

OpenCV [9], with modifications to reflect the geometry, dynamic range, and temporal characteristics of forward-facing automotive cameras. Corruption-specific generation parameters (e.g., fog airlight, rain density and streak angle, motion blur kernel size, JPEG quality factor, and exposure EV shift) are stored per-clip in the `extra_json` metadata field, enabling full reproducibility of each degraded clip.

Each corruption type is parameterized across five severity levels (1–5) using a continuous normalized severity value: $s_1 = 0.08$, $s_2 = 0.18$, $s_3 = 0.35$, $s_4 = 0.55$, $s_5 = 0.75$. Level 1 represents a mild, barely perceptible degradation; level 5 represents a severe degradation that substantially impairs scene visibility or sensor readability.

VALIDATION AND QUALITY

Capture Quality

The ten clean base clips were reviewed during and after collection to ensure sufficient visual quality across all capture conditions. Camera settings were iteratively adjusted per session to reduce common issues such as highlight clipping, low contrast in cloudy conditions, and excessive motion blur.

Nighttime captures (S06–S08) were validated to confirm that key scene elements, including lane markings, vehicles, and road structure, remained visible while preserving realistic sensor noise and low-light characteristics. Daytime captures were reviewed for sharpness, stable exposure, and representative coverage of the target environments. The dataset intentionally preserves natural acquisition variability rather than enforcing strict normalization, as this variability is relevant to robustness and reliability evaluation.

The optical and sensor characteristics of the DJI Osmo Action 4 are relevant to interpreting the clean base clips. The camera uses a 1/1.3-inch CMOS sensor with a fixed aperture of $f/2.8$, a 155° ultra-wide field of view, and supports an ISO range of 100–12800 for video. At the nighttime capture setting of ISO 1600, luminance noise is present but controlled, consistent with manufacturer characterization of the sensor; noise becomes more pronounced above ISO 1600, particularly in shadow regions [7]. The maximum video bitrate is 130 Mbps, and all clips are recorded in H.264/MP4 format prior to the downsampling step. The camera uses electronic image stabilization (EIS); all in-camera stabilization and low-light enhancement were disabled during data collection to preserve native sensor output. These characteristics define the baseline noise floor and dynamic range of the clean clips from which all corrupted variants are derived.

Anonymization Quality

Anonymization quality is recorded for each processed clip. Metadata fields include per-frame counts of detected face

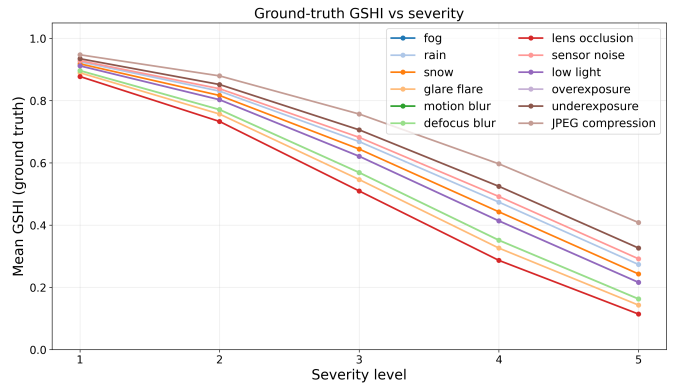


Fig. 3: Ground-truth GSHI as a function of degradation severity (levels 1–5) across all 12 corruption types. All corruptions exhibit a consistent monotonic decrease, confirming the GSHI metric is well-ordered across severity levels.

regions, license plate regions, and fallback anonymization events. This logging enables systematic auditing of privacy coverage and supports targeted inspection of challenging cases, including small faces, oblique license plates, nighttime scenes, and motion-blurred frames.

The anonymization pipeline is implemented using OpenCV [9], PyTorch [10], and Ultralytics YOLO models [8].

Near-Duplicate Reduction

The 10 clean scenarios are selected as fixed 128-frame windows from their respective source videos, chosen to avoid highly redundant segments such as stationary stops or prolonged straight-line highway sections with minimal scene change. This explicit clip-selection approach replaces a general perceptual-hash deduplication step, providing more deliberate control over scenario diversity. Perceptual hashing [11] is available in the processing codebase for use in future dataset expansions.

GSHI Definition and Ground-Truth Behavior

The Global Sensor Health Index (GSHI) [12] is a scalar metric in the range $[0, 1]$ that quantifies the perceptual impact of a degradation on a given clip. Ground-truth GSHI values (`gshi_gt`) are computed directly from the applied corruption severity and per-corruption taxonomy weights, making them fully deterministic given the generation parameters.

As shown in Fig. 3, `gshi_gt` exhibits a consistent monotonic decrease with increasing severity across all 12 corruption types. This holds across all three degradation families — weather, optical, and sensor — confirming that the GSHI formulation is sensitive to degradation intensity and well-ordered across severity levels.

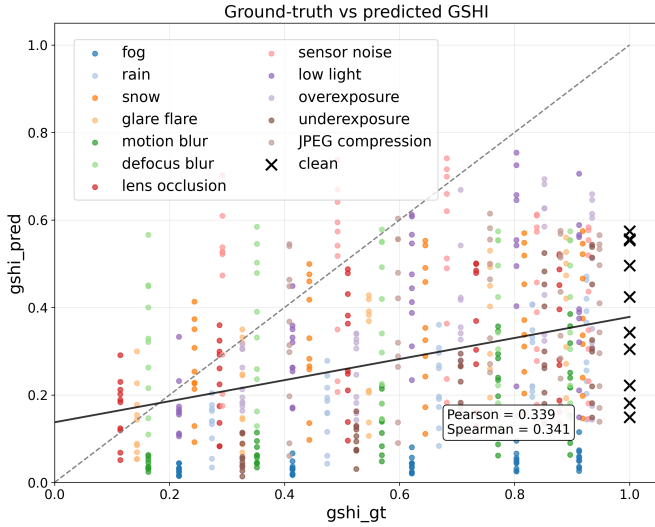


Fig. 4: Predicted versus ground-truth GSHI across all 610 clips (Pearson $r = 0.339$, Spearman $\rho = 0.341$). The model shows a systematic bias toward lower predicted values and increased dispersion at higher ground-truth GSHI levels.

Baseline Prediction Analysis

To assess the practical utility of DRIVE-C as a benchmark, we evaluate PerceptionHealthNet, a baseline model that directly predicts GSHI from input clip frames. Across all 610 clips, the Pearson correlation between predicted GSHI (`gshi_pred`) and ground-truth GSHI is $r = 0.339$, with a Spearman rank correlation of $\rho = 0.341$. The mean predicted GSHI is 0.380 for clean clips and 0.283 for corrupted clips, indicating that the model captures a coarse separation between clean and corrupted inputs, despite its limited fine-grained correlation with ground-truth GSHI.

Fig. 4 shows the scatter of predicted versus ground-truth GSHI across all samples. Predictions exhibit a systematic bias toward lower values and increased dispersion, particularly at higher ground-truth GSHI levels, suggesting the model underestimates perceptual quality for mild-to-moderate degradations.

Per-Corruption Baseline Results

Table 3 reports per-corruption Pearson and Spearman correlations, monotonicity of predicted GSHI across severity levels, and mean predicted GSHI at severities 1, 3, and 5 for all 12 corruption types.

The model performs best on underexposure ($r = 0.77$, $\rho = 0.84$) and motion blur ($r = 0.73$, $\rho = 0.80$), where predicted GSHI decreases monotonically and substantially from s_1 to s_5 . Moderate performance is observed for fog, rain, glare/flare, lens occlusion, low light, and overexposure ($r = 0.47$ – 0.64). The model fails to track severity for sensor noise ($r = -0.16$), JPEG compression ($r = 0.09$), defocus blur ($r = 0.12$), and snow ($r = 0.29$), where predictions remain nearly flat across severity levels. Only 6

of 12 corruption types show globally monotonic predicted GSHI from s_1 to s_5 (monotonicity fraction: 0.475).

These results confirm that DRIVE-C exposes substantial headroom for degradation-aware modeling, particularly for perceptually subtle corruptions that the baseline cannot reliably rank.

Known Limitations

DRIVE-C comprises 10 scenarios drawn from video collected on two recording sessions, which limits diversity in geographic coverage, vehicle type, and weather conditions. The dataset focuses on forward-facing monocular camera data and does not include synchronized lidar, radar, or stereo modalities. Dense semantic annotations such as per-frame bounding boxes or segmentation masks are not provided. The baseline model (PerceptionHealthNet) shows limited sensitivity to perceptually subtle corruptions such as sensor noise, JPEG compression, and defocus blur, which the dataset exposes as open research challenges. DRIVE-C is therefore most suitable for robustness evaluation, degradation-aware modeling, and sensor health monitoring research rather than supervised perception training requiring dense labels. Together, these results demonstrate that DRIVE-C provides a structured and challenging benchmark for advancing degradation-aware perception and sensor health estimation in autonomous driving systems.

RECORDS AND STORAGE

DRIVE-C is a video clip dataset. All clips are stored as MP4 files at 1280×720 resolution and 30 fps. The dataset comprises 610 clips in total: 10 clean clips and 600 corrupted clips (12 corruption types $\times$ 5 severity levels $\times$ 10 scenarios). Each clip is 128 frames (≈ 4.3 s). The total frame count across all clips is 78,080 (1,280 clean; 76,800 corrupted). Clips are partitioned into `dev` and `test` splits of 305 clips each, stratified across corruption types and scenarios.

Table 4 summarizes the top-level dataset statistics.

Directory Structure

The dataset is organized into two top-level directories. Clean clips are stored flat under `clean_clips/`. Corrupted clips are organized hierarchically by corruption type and then severity level under `corrupted/`, making it straightforward to load all clips of a given corruption type or a given severity across corruption types.

```
drive_c/
  clean_clips/
    S01_clean.mp4
    S02_clean.mp4
    ...
    S10_clean.mp4
```

Table 2: The 12 corruption types in DRIVE-C organized by degradation family.

Family	Corruption Type	Description
Weather	Fog	Homogeneous scattering reducing contrast and depth cues
	Rain	Streak-based precipitation with motion-dependent angle
	Snow	Particle-based accumulation with brightness variation
	Glare / flare	Specular highlight blooming from direct or reflected light
Optical	Motion blur	Linear kernel blur along the direction of vehicle motion
	Defocus blur	Radially symmetric blur simulating lens focus error
	Lens occlusion	Partial obstruction from water droplets or debris
	Low light	Luminance reduction with amplified sensor noise
Sensor	Sensor noise	Gaussian additive noise applied at the pixel level
	Overexposure	Global gain increase causing highlight clipping
	Underexposure	Global gain reduction causing shadow crushing
	JPEG compression	Block-artifact compression at decreasing quality factors

```

corrupted/
fog/
  s1/ S01_fog_s1.mp4 to S10_xx.mp4
  s2/ S01_fog_s2.mp4 to S10_xx.mp4
  ...
  s5/ S01_fog_s5.mp4 to S10_xx.mp4
rain/
  s1/ ... s5/
snow/
  s1/ ... s5/
[one subdirectory per corruption type]

```

Each corrupted clip filename follows the convention

`[scenario_id]_[corruption_type]_[severity_name].mp4` (e.g., `S03_motion_blur_s4.mp4`), enabling unambiguous identification without consulting the metadata file.

Metadata Files

All metadata is provided in two CSV files distributed with the dataset.

`final_metadata.csv` is the primary metadata file with one row per clip (610 rows, 27 fields). It covers clip identity, corruption labels, severity values, scene context, video properties, generation parameters, ground-truth GSHI, and baseline model predictions. Table 5 lists the key fields.

`scenario_metadata.csv` provides per-scenario source video details including the source video identifier, clip frame range, capture date, original resolution, and scene annotations (10 rows).

Clip Relationships

Each clean clip `SXX_clean.mp4` is the direct source for 60 corrupted clips (12 corruption types $\times$ 5 severity levels). All corrupted clips derived from the same scenario are pixel-aligned to their clean counterpart, enabling paired clean-corrupted evaluation. The `scenario_id` and `source_video_id` fields in `final_metadata.csv` provide full traceability from any corrupted clip back to its clean base and originating source video.

INSIGHTS AND NOTES

Suggested Usage Patterns

DRIVE-C supports several distinct research workflows beyond the robustness benchmarking use case described in the Validation section.

Paired clean-corrupted evaluation. Each corrupted clip has a pixel-aligned clean counterpart under the same scenario identifier. This enables direct subtraction-based analysis of degradation effects, no-reference versus full-reference quality metric comparison, and controlled ablation of model components sensitive to specific corruption families.

Severity-aware training and fine-tuning. The continuous `severity_value` field ($s_1 = 0.08$ through $s_5 = 0.75$) can be used as a regression target or curriculum signal for training degradation-aware models, beyond the five discrete severity levels.

Multi-label degradation detection. The `pred.presence.json` and `pred.severity.json` fields provide per-clip probability estimates across all 12 degradation classes, supporting multi-label classification and degradation mixture studies.

Time-series and temporal analysis. Each clip is a contiguous 128-frame sequence at 30 fps. Researchers

Table 3: Per-corruption baseline evaluation of GSHI prediction across all 12 corruption types (50 clips each). Pearson r and Spearman ρ measure correlation with ground-truth GSHI. Mono. indicates whether predicted GSHI is monotonically non-increasing from s_1 to s_5 . $\bar{p}_{s1}$, $\bar{p}_{s3}$, $\bar{p}_{s5}$ are mean predicted GSHI at each severity.

Corruption Type	Pearson r	Spearman ρ	Mono.	$\bar{p}_{s1}$	$\bar{p}_{s3}$	$\bar{p}_{s5}$
Underexposure	0.77	0.84	Yes	0.353	0.208	0.059
Motion blur	0.73	0.80	Yes	0.229	0.135	0.041
Rain	0.64	0.65	Yes	0.322	0.226	0.132
Low light	0.60	0.56	No	0.426	0.460	0.171
Glare / flare	0.56	0.55	Yes	0.355	0.286	0.135
Overexposure	0.51	0.45	No	0.434	0.368	0.230
Fog	0.50	0.55	Yes	0.057	0.041	0.028
Lens occlusion	0.47	0.45	Yes	0.341	0.309	0.167
Snow	0.29	0.27	No	0.372	0.354	0.259
Defocus blur	0.12	0.13	No	0.380	0.378	0.325
JPEG compression	0.09	0.07	No	0.372	0.386	0.333
Sensor noise	-0.16	-0.17	No	0.388	0.522	0.457

Table 4: Top-level DRIVE-C dataset statistics.

Property	Value
Total clips	610
Clean clips	10
Corrupted clips	600
Scenarios	10 (S01–S10)
Corruption types	12
Severity levels	5 (s_1 – s_5)
Frames per clip	128
Total frames	78,080
Resolution	1280×720 (HD)
Native resolution	3840×2160 (4K UHD)
Frame rate	30 fps
Format	MP4 (H.264)
Dev / test split	305 / 305 clips
Capture dates	April 1 and April 5, 2026
Capture location	Michigan, United States

studying temporal dynamics of degradation effects, optical flow robustness, or video quality assessment can exploit the temporal structure rather than treating clips as independent frame collections.

General video quality assessment. Although DRIVE-C is designed for automotive perception, the controlled degradation structure and GSHI metric are not specific to driving. The dataset can serve as a benchmark for no-reference video quality estimators in any domain where camera sensor health is relevant.

Caveats and Special Cases

Scene type imbalance. The 10 scenarios are not uniformly distributed across scene types: 6 are urban, 2 rural, 1 freeway, and 1 parking lot. Urban scenarios are over-represented, and scene types such as tunnels, construction zones, and highway on-ramps are absent. Researchers

drawing conclusions about freeway or parking-lot-specific robustness should account for this imbalance.

Nighttime scenario concentration. All three nighttime scenarios (S06, S07, S08) are urban and cloudy. There are no nighttime rural, freeway, or clear-sky scenarios in the current release. Models evaluated on nighttime clips in DRIVE-C are therefore evaluated specifically on urban nighttime conditions, not nighttime driving in general.

Baseline model false positives on clean clips. The baseline PerceptionHealthNet model assigns low predicted GSHI to several clean clips, with nighttime scenarios particularly affected: S08 ($\hat{g} = 0.149$, predicted class: vignetting) and S07 ($\hat{g} = 0.181$, predicted class: vignetting) receive lower health scores than many corrupted clips. This reflects the model’s difficulty distinguishing natural low-light acquisition characteristics from synthetic degradation. Users relying on `gshi_pred` for quality filtering should validate thresholds against the clean clip distribution before applying them to new data.

Non-monotonic baseline predictions for subtle corruptions. For sensor noise, JPEG compression, defocus blur, and snow, the baseline model’s predicted GSHI does not decrease monotonically with increasing severity — and for sensor noise the Pearson correlation with ground-truth GSHI is negative ($r = -0.16$). These corruption types should be treated as open benchmarking challenges rather than solved cases in any downstream evaluation.

Dev/test split assignment. Scenarios S01–S05 are assigned to the `dev` split and S06–S10 to the `test` split. Because split assignment is at the scenario level (not the clip level), the dev and test sets differ not only in sample identity but also in scene type and lighting distribution: the dev set contains the freeway and parking-lot scenarios, while the test set contains all three nighttime scenarios. Researchers should be aware of this structural difference when interpreting generalization results between splits.

Table 5: Key fields in `final_metadata.csv` (610 rows, 27 fields).

Field	Type	Description
<code>sample_id</code>	String	Unique clip identifier, e.g., <code>S01_fog_s3</code> .
<code>scenario_id</code>	String	Scenario identifier (<code>S01</code> – <code>S10</code>).
<code>source_video_id</code>	String	Source video identifier for traceability.
<code>split</code>	String	Benchmark split: <code>dev</code> or <code>test</code> .
<code>clip_type</code>	String	<code>clean</code> or <code>corrupted</code> .
<code>corruption_type</code>	String	Corruption name, e.g., <code>fog</code> , <code>motion_blur</code> .
<code>severity_level</code>	Integer	Integer severity (0 = clean, 1–5 = corrupted).
<code>severity_name</code>	String	Severity label, e.g., <code>s0</code> , <code>s1</code> , ..., <code>s5</code> .
<code>severity_value</code>	Float	Continuous severity used by the generator.
<code>is_clean</code>	Boolean	Indicates whether the sample is an uncorrupted clean clip.
<code>gshi</code>	Float	Assigned Global Sensor Health Index value for the clip.
<code>weather</code>	String	Scene weather, e.g., <code>sunny</code> or <code>cloudy</code> .
<code>time_of_day</code>	String	Lighting regime: <code>day</code> or <code>night</code> .
<code>scene_type</code>	String	Scene label: <code>urban</code> , <code>rural</code> , <code>freeway</code> , etc.
<code>traffic_level</code>	String	Approximate traffic density, e.g., <code>none</code> , <code>low</code> , or <code>moderate</code> .
<code>fps</code>	Integer	Frame rate (30 for all clips).
<code>resolution</code>	String	Clip resolution (1280x720 for all clips).
<code>num_frames</code>	Integer	Number of frames (128 for all clips).
<code>output_path</code>	String	Relative path to the MP4 file within the dataset.
<code>extra_json</code>	JSON	Per-clip corruption-generation parameters for reproducibility.
<code>gshi_gt</code>	Float	Ground-truth GSHI computed from severity and taxonomy weights.
<code>gshi_pred</code>	Float	Baseline model predicted GSHI.
<code>pred_top1_issue</code>	String	Highest-probability predicted degradation class.
<code>pred_top1_prob</code>	Float	Probability of the top predicted class.
<code>pred_presence_json</code>	JSON	Predicted presence probabilities for all degradation classes.
<code>pred_severity_json</code>	JSON	Predicted severity values for all degradation classes.
<code>pred_present_thresh_json</code>	JSON	Thresholded predicted-present flags for degradation classes.

Privacy processing and native sensor characteristics. Corrupted clips are generated from anonymized clean clips. The anonymization step (localized face and license plate blurring) modifies small regions of each frame. For the vast majority of pixels and for global image statistics, this has no effect. However, studies specifically examining face or vehicle detection robustness under degradation should note that face and plate regions have been replaced with blurred content.

Corruption independence assumption. Each corrupted clip contains exactly one active corruption type. Real-world sensor degradation often involves simultaneous or correlated degradations (e.g., rain combined with lens occlusion, or low light combined with motion blur). DRIVE-C’s single-corruption design enables controlled evaluation but does not directly cover compound degradation scenarios. Researchers interested in compound degradations can construct them by applying the provided generation scripts to the clean clips with multiple active corruption types.

SOURCE CODE AND SCRIPTS

The DRIVE-C dataset is constructed using a Python-based processing pipeline. The associated repository pro-

vides utilities for:

- selecting and extracting scenario clip windows from source videos,
- anonymizing faces and license plates using GPU-accelerated YOLO-family detectors,
- generating controlled corruption variants across all 12 corruption types and 5 severity levels,
- computing ground-truth GSHI values from applied severity and taxonomy weights,
- running the PerceptionHealthNet baseline model to produce `gshi_pred` and per-class degradation predictions, and
- generating and validating the `final_metadata.csv` and `scenario_metadata.csv` manifest files.

The codebase is publicly available at:

<https://github.com/shiv-aher/drive-c-dataset>

The repository includes all scripts and configuration files required to reproduce the dataset generation process from the original source videos. Corruption-specific

generation parameters for each clip are stored in the `extra_json` field of `final_metadata.csv`, enabling any individual corrupted clip to be regenerated independently without rerunning the full pipeline.

To support long-term reproducibility, the source code repository will be archived with a persistent DOI via Zenodo corresponding to the dataset release version. The dataset itself will be publicly released through a permanent data repository with open access.

The implementation is written in Python and leverages widely used open-source libraries including OpenCV [9], PyTorch [10], and Ultralytics YOLO [8].

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